Looped Transformers and Recurrent Depth

Eight years from Universal Transformers to GPT-6 Astra (2018–2026): five core papers and ten trend insights

Presenter: awesome_loop_transformer project

github.com/asimfish/awesome_loop_transformer

First built 2026-09-06 · paper list updated 2026-11-06: 333 papers (211 added)

Outline

1. Why now
2. Five core papers
3. The wider map, ten insights, and what changed since September
4. Open problems and takeaways

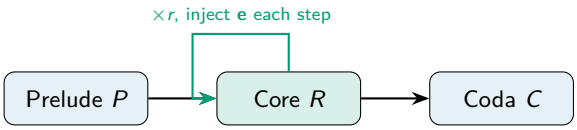
A 3.5B model that spends FLOPs like a 32B one

Reported, September 2026: The Information says GPT-6 Astra uses "recurrent depth".

The idea

Apply **one set of parameters** repeatedly along depth:

$$\mathbf{s}_i = R(\mathbf{e}, \mathbf{s}_{i-1}), \quad i = 1, \ldots, r$$



Source: Geiping et al. 2025, Fig. 2 (redrawn)

- Depth $2 + 4r + 2$ grows with r; parameters stay at 3.5B
- r=4 → 32: ARC-E 49.1 → 69.9, GSM8K-CoT ≈ 0 → 42.1 (Huginn)
- No chain-of-thought data; small context window

Takeaway: recurrence trades parameters for compute; this talk asks when that trade wins.

Taxonomy: where the loop sits, and three orthogonal axes

Granularity	Definition	Representatives
model-loop	whole stack fed back	UT (2018), Huginn core, Ouro
block-loop	contiguous middle segment loops	Huginn 3-stage, Hyperloop, Training-Free, SMELT
layer-loop	each layer repeats K times, then next layer	Loopie, Training-Free layer-mode, MixerLoop
latent conditioning	previous state injected as condition	Coconut, PLT (CLP), Latent Recurrent Tr.

Loop count
fixed (Loopie, LoopCoder-v2) / random-train, free-test (Huginn) / per-token halting (ACT, MoR, TaH)

Training
from scratch (Huginn, Ouro, Loopie) / retrofit (Relaxed Recursive, Retrofitted Recurrence) / training-free

What the loop is
unrolled LM / fixed point (DEQ, STARS) / ODE refinement (Training-Free) / supervised latent CoT (LOTUS) / RL on steps (RLTT)

Universal Transformers (2018): recurrence + adaptive halting

Mechanism (Dehghani et al., ICLR 2019)

$$A^t = \mathrm{LN}\big(H^{t-1} + P^t + \mathrm{MHSA}(H^{t-1} + P^t)\big)$$
$$H^t = \mathrm{LN}\big(A^t + \mathrm{Transition}(A^t)\big)$$

- Parameters shared over time steps t; coordinate embedding P^t = position + time step
- ACT halting per position: ponder time 2.3 / 3.1 / 3.8 steps for 1 / 2 / 3 supporting facts
- Turing-complete given memory (reduces to Neural GPU)

Task	Trans.	UT	+ACT
bAbI 10K joint (err%)	22.1	0.47	**0.29**
LAMBADA ppl	7321	319	**142**
WMT14 En-De BLEU	28.0	**28.9**	–

Open problems it left: parameter–compute ratio, training stability, inference cost.

Takeaway: gains concentrate on hard instances; the same pattern reappears in Huginn seven years later.

Huginn-3.5B (Feb 2025): recurrent depth trained at modern scale

- Shape $(2, 4, 2)$, $h = 5280$, 3.5B params; $r = 32 \Rightarrow 132$ effective layers
- Train: $r \sim$ log-normal Poisson ($\bar{r} = 32$), backprop through last $k = 8$ steps only
- 800B tokens, 4096 AMD MI250X, constant LR 4×10^{-5}, sandwich norm
- Two failed runs: hidden-state collapse; recurrence collapse (r=1 ≡ r=32)

Fair comparison is at 0.18T tokens (Table 4)

Same setup, non-recurrent: ARC-E 46.42, GSM8K-CoT 2.20 vs. recurrent r=32: **53.62**, **10.24**

r	ARC-E	HellaSwag	MMLU	GSM8K CoT
4	49.07	43.46	23.39	–
8	65.11	58.54	25.29	–
16	69.49	64.67	31.25	–
32	**69.91**	**65.21**	**31.38**	**42.08**

Table 1–2; GSM8K CoT strict, with system prompt.

The last 620B tokens improve r=32 but not r=1 (34.01 → 34.89): everything learned lives in the loop.

Huginn: what the loop buys, and what emerges in latent space

Zero-shot capabilities

- Per-token adaptive exit (KL $< 5\times10^{-4}$): MT-Bench 5.63 $\rightarrow$ 5.56
- KV-cache sharing with budget 4: no loss
- Continuous CoT: warm-start $\mathbf{s}_0$ from previous token
- Self-speculative decoding without a draft model

- More context $\Rightarrow$ more loops needed: ARC-C saturates at 8 (0-shot) vs. 32 (25–50-shot)
- Open-book OBQA +11.0: reasons over context, memorizes little

Latent geometry (Sec. 7)

Fixed points for most tokens; **orbits** for numeric tokens; **sliders** (steady drift) for key tokens; path independence across $\mathbf{s}_0$.

Caveats: one run, LR never cooled; r=32 inference is 16$\times$ an 8-layer model; Lu et al. 2025 find limited interpretable latent CoT.

Takeaway: "3.5B behaves like 50B" is FLOP-equivalence, not benchmark-equivalence.

Training-Free Looped Transformers (May 2026): a layer is one Euler step

Insight (Chen, Li, Liang, Lao, Liu; UT Austin)

Pre-norm layer $L(x) = x + F(x)$ is forward Euler (h=1) on $\dot{x} = F(x)$. Naive K-fold looping integrates to t=K — post-loop layers never saw that.
Sub-step instead:

$$x_{k+1} = \left(1 - \tfrac{1}{K}\right) x_k + \tfrac{1}{K}\, g(x_k)$$

- RK tableau with anchor β: output $= \beta\, g(x_0) + (1 - \beta)\, F^K(x_0)$
- Toy net, K=8: naive MSE 335 vs. sub-step 1.04; accelerators (Anderson, heavy-ball, Aitken) never beat RK $\Rightarrow$ mid-stack is an ODE flow, not a contraction

Model	Benchmark	Base	Δpp
Qwen3-4B-Instruct	MMLU-Pro	57.1	**+2.64**
Qwen3-4B-Instruct	GPQA-Main	33.7	**+2.01**
Qwen1.5-MoE-A2.7B	ARC-C	48.3	**+2.30**
Llama-3.2-1B-Inst.	GPQA-Main	27.9	+1.79
Moonlight-16B-A3B	OBQA	31.6	+1.20
DeepSeek-V2-Lite	ARC-C	57.9	+0.85

One recipe, no per-cell tuning: middle 4 layers, 3-stage RK, block-mode dense / layer-mode MoE.

Training-Free: window size, depth fraction, and the MoE rule

Window n	Layers	Δ (16-task)
1	[14]	+0.18
3	[13–15]	+0.21
4	[12–15]	**+0.55**
6	[11–16]	−0.82
28	all	−27.73

Qwen3-1.7B-Base, forward Euler K=2 (Table 2a).

- **Depth-fraction rule**: optimal window center at 0.43–0.71 of depth (>1.7B), 0.25–0.56 (<1B)
- **MoE needs layer-mode**: block-mode re-routes experts each iteration (routing thrash); layer-mode gates once
- 45 cells: 60% positive, 27% neutral, 87% non-negative; failures = sub-3B distilled on knowledge MC
- Naive loop at K=6: −17.71pp; this method stable for $K \leq 24$

Takeaway: an untrained model can only be looped in a small mid-stack window; gains cap at ∼2–3 points.

LoopCoder-v2 (Jun 2026): in a Parallel Loop Transformer, loop 2 is almost everything

PLT (Wu et al. 2025) flattens cost

- Shared first-loop KV + gated sliding window (w=64): KV $O(LSd)$
- Cross-loop position offset: $B^{(r)} = \mathrm{Embed}(x) + \mathrm{shift}(\mathbf{h}^{(r-1)}) \Rightarrow$ loops run in one pass
- Price: token i receives its neighbor's state (positional mismatch)

7B	HE+	LCB	SWE-V	Multi-SWE	Avg
R=1	81.1	27.4	43.0	14.0	38.0
R=2	**84.1**	**35.4**	**64.4**	**31.0**	**46.5**
R=3	75.0	28.6	27.6	11.0	36.9
R=4	76.8	24.5	22.4	9.3	34.3

Table 2. R=2 on SWE-bench Verified (64.4) beats Kimi-Dev-72B (60.4) and Qwen3-235B-A22B (45.2).

7B dense, $R \in \{1, 2, 3, 4\}$ each trained from scratch on 18T tokens; same SFT; 1M GPU hours.

Takeaway: strongly non-monotonic in R — the next slide shows why.

LoopCoder-v2: gain–cost scissors, and CoT + loop is super-additive

Loop	$\delta^{(r)}$	$\Delta p^{(r)}$	eff. rank	$\cos\theta$
r=2	**846**	**1.75**	**174.6**	−0.72
r=3	464	1.32	172.5	−0.46
r=4	1014	1.58	158.2	0.04

PLT4, 500 samples (Table 3). Offset cost $\Omega^{(r)} = \frac{1}{S}\sum_i \|\mathbf{h}_i^{(r-1)} - \mathbf{h}_{i-1}^{(r-1)}\|$ is $\approx$ constant.

- Effective rank peaks at loop 2, then narrows
- Updates oscillate ($\cos\theta < 0$); attention routing freezes after loop 2
- Fixed offset tax is **30–45**$\times$ the marginal gain from loop 3 on

Thinking mode on top of R=2 (Table 4)

	Instruct	Thinking
LiveCodeBench	35.4	**62.3**
CRUX	86.9	**93.5**
MultiPL-E	73.9	**77.8**

Explicit CoT alone does not help the non-looped model; loop alone gives single digits. Together: super-additive.

Scope: "two is optimal" is a PLT property (CLP tax). Huginn improves monotonically to r=32.

Loop the Loopies! (Jul 2026): the first compute-matched win

Loopie Recipe (match by measured step time)

1. Halve stored layers, loop twice: LR unchanged, activation memory $\propto sbDL$ halves
2. Spend memory on $2\times$ micro-batch (tokens/step unchanged)
3. Reinvest measured speed-up into width; pick equal wall-clock step time

Config	D	L	R	$\hat{C}$	$\hat{M}_{\text{act}}$
Qwen3-30B-A3B ref.	2048	48	1	1.00	1.00
seed Loopie	2048	24	2	1.00	0.50
Loopie-20B-A2B	2304	27	2	**1.42**	0.63

- **Layer-loop** ($L_1L_1L_2L_2\ldots$) overtakes model-loop after $\sim$1.2T tokens; pipeline-parallel friendly
- Compute-matched run: overtakes reproduced Qwen3-30B-A3B after $\sim$600B of 800B tokens
- Scaling ladder 0.15B–1B active params: Loopie wins all 4 rungs, gap does not shrink
- Why R=2: marginal return of $N\times$ loops vs. $N\times$ layers decays fastest after 2

Accounting: nominal FLOPs +42%; inference does 54 vs. 48 block passes per token; no latency reported.

Loopie: post-training with supervised pre-training (SPT) and RL

SPT = SFT loss mask + pre-training scale

Loss on target tokens only; global batch $\geq$1024, sequence $\geq$128K, $\geq$128M token positions per batch, 2T tokens, 10 epochs. Pretraining metrics $\uparrow$ and reasoning metrics $\uparrow$; no overfitting, no forgetting.

RL: GSPO (sequence-level ratio/clip) + DAPO clip-higher and dynamic sampling; math then code; 32K $\rightarrow$ 64K.

	Qwen3 30B-A3B (36T)	Nemotron 3 Nano (25T)	**Loopie 20B-A2B (3.5T)**
MMLU	85.83	80.52	**81.28**
ARC-C	96.67	91.96	93.52
BBH	86.59	68.76	82.28
IFEval	85.58	77.05	84.72
AIME 24	90.10	85.00	**92.09**
AMC	96.73	91.80	94.21

Table 3. Loopie-6B-A0.6B Thinking: AIME24 80.42, AIME25 70.83 (Ouro-2.6B: 62.50 / 51.67).

Takeaway: under one-seventh of the pretraining tokens, a looped MoE matches 25T-token peers on knowledge benchmarks.

The 2026 literature: ten themes, roughly sixty papers

Theme	Representative results
Theory	k layers looped L times $\approx kL$ layers (Saunshi 25); loops vs. CoT separation (Xu & Sato 25); budget law $v \approx n/T$ (Zhang 26)
Scaled pretraining	Huginn 3.5B/0.8T, Ouro 2.6B/7.7T, LoopCoder-v2 7B/18T, Loopie 20B-A2B/3.5T, Nanbeige4.2-3B/28T, SMELT to 54B; loop counts 2–4
Efficiency	PLT (latency), diffusion-style sampler 5×, MELT constant KV, Looped Latent Attention 32× KV, LT2 linear attention, LoopQ
Stability	residual scaling $\varepsilon{=}1/N$ (not $1/\sqrt{L}$), DeepLoop exponent $1/4 \to 1/2$, STARS spectral radius, settling/drifting regimes
Adaptive halting	MoR (NeurIPS 25), Think-at-Hard (skip 93% of tokens), LoopFormer, RecurTrace (2.0 loops avg.)
Retrofit / training-free	Relaxed Recursive (LoRA), Retrofitted Recurrence, Training-Free RK, Program-of-Layers (ICML 26)
Post-training	RLTT trajectory credit +10.9%, LoopRPT, LOTUS latent CoT matches explicit at 3B, SPT
Interpretability	orbits/sliders (Huginn), limited latent-CoT evidence (Lu 25), Jacobian lens: Ouro rebuilds workspace per loop
Tiny reasoners	HRM 27M, TRM 7M (ARC-AGI-1 45%), Attractor Models, Denoising Recursion
Applications	LoopViT (ARC 65.8% at 18M), RD-VLA (0% → 90%, 80× faster), compositional tool calling, agentic products

Insights 1–3: narrow win, two regimes, converging on the middle

1. **Compute-matched wins exist, through a narrow door.** Loopie (step time) and SMELT (FLOPs/params/KV, 6.8–18% savings) win only with MoE + layer/mid-stack loop + two loops. One recurrence is worth $r^{0.46}$ unique parameters (0.38 under truncated backprop): loops cannot replace parameters; Loopie converts *activation memory* into width.
2. **"How many loops" splits by where compute binds.** Fixed pretraining budget $\Rightarrow R{=}2$ (Loopie, LoopCoder-v2, SMELT, Nanbeige). Variable inference compute $\Rightarrow$ adaptive and deep (Huginn: 8 loops zero-shot, 32 with 25–50 shots). "Three regress" is PLT's positional tax.
3. **Granularity shrinks, placement converges.** Whole model $\rightarrow$ three-stage $\rightarrow$ per-layer / mid-stack $\rightarrow$ mixer-only (MixerLoop: 41.5% of gain at -45.9% looped FLOPs). Optimal window center ≈ 0.5 of depth; MoE needs layer-mode (two independent papers agree).

Insights 4–7: ODE vs. fixed point, stability theorems, inference cost, CoT complementarity

4. **Two philosophies.** Training-Free: a pretrained mid-stack is an ODE flow (sub-step, keep the endpoint). STARS / Fixed-Point Reasoners / Attractor Models: *train* the loop to be a fixed point. Viakhirev et al.: settling operators are depth-safe; Huginn is non-settling; LoopCoder-v2's $\cos\theta < 0$ is drift.
5. **Stability went from folklore to theorems.** $\varepsilon{=}1/N$ residual scaling; DeepNorm exponent $1/2$; LR depends only on unique layers and transfers across N; stable training to 12–15 loops.
6. **Inference cost decomposes into three stackable fixes.** Latency (PLT, diffusion sampler $5\times$), KV (sharing; LLA low-rank $32\times$), quantization (LoopQ). Matched training compute $\neq$ matched inference cost.
7. **Loops and CoT are complements.** LoopCoder-v2 LCB $+26.9$ super-additive; LOTUS matches explicit CoT at 3B with 2.5–6.9$\times$ lower thought latency. Standard GRPO credits only the final latent state: RLTT ($+10.9\%$) and LoopRPT redesign credit assignment.

Insights 8–10: capability profile, applications, productization

8. **Loops manipulate knowledge; they do not store it.** Ouro attributes gains to knowledge manipulation; Huginn open-book +11.0; Frey et al.: loops help math, memory banks help commonsense; Dual-Path gates send arithmetic tokens "deep" and function words "wide". Best partners: retrieval, long context, MoE expert banks.
9. **From puzzles to agents and embodiment.** SWE-bench Verified +21.4 (LoopCoder-v2); compositional tool calling improves with depth; Nanbeige4.2-3B beats Qwen3.5-9B on agentic benchmarks; RD-VLA 0% $\rightarrow$ 90% with four iterations, 80$\times$ faster than token-reasoning VLAs.
10. **Productization started; tooling lags.** Five open families (Huginn, Ouro, LoopCoder-v2, Loopie, Nanbeige4.2-3B). Nanbeige4.2-3B ships with five out-of-the-box bugs in HF transformers and 2$\times$ attention memory from layer reuse. No standard interface yet for looped inference, KV, or quantization.

What changed since September 2026

211 papers added on 2026-11-06: 167 first submitted 2026-09-01 to 2026-11-06, plus 44 June–August papers the first search missed.

#	Change	Key number (conditions)	arXiv	Insight
1	**From-scratch loops beat matched baselines without MoE**; retrofitting is a mixed bet	Loopback: lower loss with **40–75%** of the Transformer parameters (matched tokens, executed depth)	2611.07226	refines 1
2	**Two loops is the optimum of current designs, not a rule**	LOOM 700M: best at **5 loops**, near-iso-FLOPs	2610.01153	revises 2
3	**Serving is the largest cluster**; looped models run in production engines	LoopVQ in SGLang: up to **6.1**× Ouro serving throughput	2611.05304	revises 10; refines 6
4	**Loop depth changes safety behaviour**	Ouro-1.4B-Thinking, half depth: harmful-help rate up **5.3**×, 89% of GSM8K accuracy kept	2610.15218	new; cf. 7
5	**"Manipulate, not store" becomes architecture**	Fact edits propagate **0.71–0.96** with memory vs. 0.04–0.30 with loops alone (synthetic)	2606.20737	supports 8
6	**Applications reach generative and scientific models**	260M Looped-DiT beats a 6.5× larger text-to-image model at **4.9**× lower inference compute	2609.40305	extends 9

Takeaway: insight 2 and the toolchain half of 10 are revised; post-training must cover every depth.

Five circulation errors, checked against the papers

Common claim	What the paper says
Huginn crushes its baseline "at the same 0.8T tokens"	Baseline trained 0.18T; at 0.18T recurrent ARC-E 53.62 vs. 46.42 (Table 4)
"Two loops suffice, three regress" is a law	PLT-specific (CLP tax), each R trained separately; Huginn monotone to $r{=}32$
Loopie and Qwen3-30B-A3B use "the same compute"	Same measured step time; nominal FLOPs +42%; inference cost unreported; reference is an 800B-token reproduction
Training-Free makes open models "stronger"	+1–2.6pp on knowledge MC; sub-3B distilled models can lose; +15–30% latency
Huginn "performs like 50B at 3.5B"	FLOP-equivalent; on benchmarks near OLMo-7B, behind OLMo-2-7B

GPT-6 Astra: what is verifiable and what is inference

Not verifiable here

Which loop Astra uses, how many, how trained. OpenAI's page discloses no architecture; The Information's report is anonymously sourced; neither was accessible to us.

- **Inference 1.** The public recipe closest to "frontier scale + compute-matched" is Loopie/SMELT: MoE, mid-stack or per-layer, small loop count.
- **Inference 2.** "OpenAI beat two-loop saturation" starts from a false premise — saturation is PLT-specific; the real bottleneck is inference cost, already cut $\sim 10\times$ by public KV compression and parallel sampling.
- **Inference 3.** Reported "security concerns" match a real gap: latent reasoning has no readable intermediate tokens; probes depend on layer and decoding method.
- **Inference 4.** Loopie (Jul), LLA (Jul), SMELT (1 Sep), Astra (early Sep): several groups reached usability simultaneously.

Open problems

1. Compute-matched evidence for $R \geq 4$ (all wins are at R=2)
2. The full inference bill: step time + latency + peak KV in one paper
3. Scaling-law form at 100B+ (SMELT to 54B, Schwethelm to $\sim$1B)
4. Adaptive halting beyond 8B (only Ouro's entropy allocation at 7.7T)
5. Supervisability of latent reasoning without gold CoT (safety)
6. Mechanism of layer-loop (only SMELT's attention-sink account)
7. Loops vs. sequence-length scaling (Hidden Decoding, WeLM 617B) at 100B+

Most valuable gaps

Compute-matched scaling laws at $R \geq 4$; loop-specific RL objectives with trajectory-level credit.

Three takeaways

1. **Recurrent depth passed the compute-matching gate in 2026** — via layer-loop, MoE, and two loops, with the win coming from activation memory, not from loops replacing parameters.
2. **Pick the loop count by where compute binds**: 2 for frontier pretraining; adaptive and deep when inference compute is variable. Two-loop saturation is PLT's price, not a ceiling.
3. **Loop under CoT, not instead of it**; redesign RL credit for latent trajectories; measure latency and KV before shipping.

Practical recipe

Zero-training: middle 4 layers, 3-stage RK, β mid, layer-mode on MoE. From scratch: MoE, layer-loop, R=2, match by measured step time, $1/N$ residual scaling, SPT + trajectory RL. Variable inference: sequential loop, random-r training, truncated backprop, adaptive exit, KV codec.

References I

- M. Dehghani, S. Gouws, O. Vinyals, J. Uszkoreit, Ł. Kaiser. Universal Transformers. ICLR 2019. arXiv:1807.03819.
- J. Geiping, S. McLeish, N. Jain, J. Kirchenbauer, S. Singh, B. Bartoldson, B. Kailkhura, A. Bhatele, T. Goldstein. Scaling up Test-Time Compute with Latent Reasoning: A Recurrent Depth Approach. arXiv:2502.05171, 2025.
- L. Chen, J. Li, C. Liang, N. Lao, Q. Liu. Training-Free Looped Transformers. arXiv:2605.23872, 2026.
- J. Yang et al. LoopCoder-v2: Only Loop Once for Efficient Test-Time Computation Scaling. arXiv:2606.18023, 2026.
- Z. Gao, Y. Chen, Y. Xiao, X. Yang, R. Tao, J. Zhou, B. Dai. Loop the Loopies! arXiv:2607.16051, 2026.
- S. Wang et al. SMELT: Scaling Laws for Compute-Matched MoE Looped Transformers. arXiv:2609.01343, 2026.
- B. Wu et al. Parallel Loop Transformer for Efficient Test-Time Computation Scaling. arXiv:2510.24824, 2025.

References II

- R.-J. Zhu et al. Scaling Latent Reasoning via Looped Language Models (Ouro). arXiv:2510.25741, 2025.
- K. Schwethelm, D. Rueckert, G. Kaissis. How Much Is One Recurrence Worth? Iso-Depth Scaling Laws for Looped Language Models. arXiv:2604.21106, 2026.
- J. O'Neill, F. Reid. Looped Latent Attention: Cross-Loop KV Compression for Looped Transformers. arXiv:2607.15456, 2026.
- Y. Fan, A. Svete, K. Lee. Bridging the Gap Between Latent and Explicit Reasoning with Looped Transformers (LOTUS). arXiv:2606.31779, 2026.
- J. Williams, E. Tureci. Rewarding Latent Thought Trajectories Improves Reasoning in Looped Language Models. ICML 2026. arXiv:2602.10520.

Thank you

Full reports (ZH/EN), translated PDFs, HTML deck and paper list:
`https://github.com/asimfish/awesome_loop_transformer`

Backup Slides

Backup: Huginn training details

- Data: 800B tokens (795B realized), public, skewed to code/math; instruction data mixed into pretraining; vocab 65536; sequence 4096
- Hardware: ORNL Frontier, 4096 MI250X, 21 segments $\leq$12h; 108.75 TFLOP/s per node (87% achievable), 52–64 TFLOP/s per GPU at scale; data parallel + optimizer sharding only
- Optimizer: Adam (0.9, 0.95), constant LR 4×10^{-5} after 4096 warm-up steps, no decay; locked-step sampling of r per micro-batch
- Init: Takase et al. 2024, variance $2/(5h)$; output projections $1/(5hl)$ with $l = 132$
- Norm: four RMSNorms per layer ("sandwich"); RoPE base 50000; gated SiLU MLP; QK biases
- EMA over last 75 checkpoints: GSM8K flexible 47.23% at r=64

Backup: Training-Free RK tableau and cost

$$h = 1,\ s = K,\ a_{ij} = \tfrac{1}{K}\,\mathbb{I}(j < i), \qquad b_1 = \beta + \tfrac{1-\beta}{K},\ b_i = \tfrac{1-\beta}{K}\ (i \geq 2)$$

$$x_1 = x_0 + \sum_{i=1}^{K} b_i k_i = \beta\, g(x_0) + (1-\beta)\, F^K(x_0), \qquad F(x) = x + \tfrac{1}{K} F_g(x)$$

- Extra cost: $(b - a + 1)(K - 1)$ layer forwards; K=3 on a 4-layer window $\approx$ 8 extra layers (15–30% latency on 28–48-layer models)
- Seven strategies compared: naive, forward Euler, midpoint/Heun, RK4, heavy-ball, Anderson, Aitken, Uniform Loop
- Leakage-free transfer to Qwen3-30B-A3B at window $[22, 24]$: 8 of 9 benchmarks improve (CSQA $+1.14$, MMLU-flex $+0.79$)

Backup: Loopie pretraining and RL details

- Stage 1: Nemotron-CC-v2-HQ, $\sim$570B unique tokens, 4 epochs (2.28T); Stage 2: 1.26T annealing (SFT-v1 27.8%, Specialized-v1 21.9%, Code-v2 20.7%, HQ-Synthetic 15.6%, Math-v1 10.0%)
- Qwen3 tokenizer; AdamW ϵ=10^{-15}, wd 0.1, LR 5×10^{-4} (6B) / 3×10^{-4} (20B), warm-up-stable, batch 1024×8192
- Init: embeddings $\mathcal{N}(0, d^{-1})$; block parameters SmallInit $\mathcal{N}(0, 2/(5d))$
- RL: GSPO sequence-level ratio $s_i(\theta) = (\pi_\theta / \pi_{\theta_{\text{old}}})^{1/|o_i|}$ with asymmetric clipping; prompt-level dynamic filtering; Guru-RL math and code pools; 32K then 64K response length
- Code released as megatron-loopie and vllm-loopie forks